Zoology Legends

Zoo404

Current Final Term Papers 2023

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Quiz Final Term 2023

1) Donnan Equilibrium Discovered by _____ Frederick **Donnan in 1911.**
2) Thin filaments also have a three polypeptide complex troponin at intervals of about __ along the thin filament. **40nm**
3) Thyrotropin-Releasing hormone, TRH, Has only **3 amino acids**
4) Endothelia—a peptide, is a potent inducer of **Vasoconstriction**
5) In amniotes, located in superior posterolateral wall of the right atrium (a vestigial remnant of **sinus venosus).**
6) Gas exchange happens __**Passively** __by diffusion
7) Medial side deeply concave—forms a medial Depression: ___**Hilus (Hilum)**
8) Urine formation Involves__ **three**_ main Processes:
9) **Resting Tidal Volume** ___Amount of air that Moves into or out of Lungs during a Single, quiet Respiratory

Zoology Legends

10) **Hypoxia** ___• "A condition in Which the body Or a region of the Body is deprived Of adequate Oxygen supply

11) Main nutritional part of the nerve cell_____ **Cell Body**

12) The membrane of the cell is called **sarcolemma**. Its cytoplasm is called **sarcoplasm**.

13) One of the strongest pieces of evidence in support of the **sliding filament** theory comes from the length-tension relation of a sarcomere

14) Troponin complex binds to tropomyosin about every __ along the thin actin filament. **40 nm**

15) A **fused tetanic** contraction is the maximal possible contraction.

16) Physiological Causes of Muscle Fatigue, Ion imbalances occur due to inadequate functioning of the **Na+/K+ pump.**

17) Oxytocin containing **nine** amino acid residues.

18) In heat cell Na+ flux from fast sodium channels present develops a stable manageable **ingredients.**

19) Parasympathetic cholinergic fibers of **Vagus nerve (10th cranial nerve)** innervate S-A and A-V nodes.

20) Kidney Receive remarkably large blood flow: **20%- 25%** of total cardiac output

21) **Glomerular filtrate** has a chemical composition very similar to that **of blood plasma**

22) Filtration surface of each kidney is about **6 m2.**

23) Major nitrogenous waste is **amino group (–NH2).**

24) Poiseuille's Equation For Blood Flow In this equation: • ρ = density of blood • **η = viscosity of blood.**

25) Most mammals don't have **uricase, allantoicase and allantoinase** enzymes.

26) Uricotelic Animals which excrete nitrogen chiefly as **uric acid** or closely related compound **guanine.**

27) Once food reaches back of buccal cavity, swallowing becomes involuntary—under direct neural control of **medulla oblongata** of the brain

28) Motility of the Alimentary Canal prevents scumming **(layer formation).**

29) Respiratory surface area of human lung is **50 to 100 m2**

30) Average cardiac output for resting adult human: **5 L/min.**

31) Thoracic Cavity in Reptiles • Reptiles do not possess a **diaphragm**

32) Resting Tidal Volume • Amount of air that moves into or out of lungs during a single, quiet **respiratory cycle** • Averages **about 500 mL** in human male and female both

33) **Kymograph** • "A device that graphically records changes in the mechanical activities of animal tissues"

34) Lungs of Birds relatively small Connected to **7 or 9** air sacs

35) A condition in which the body or a region of the body is deprived of adequate oxygen supply - --------- **Hypoxia**

36) Urine formation involves **three** main processes

37) Medial side of mammalian kidney deeply concave—forms a medial depression: **Hilus (Hilum)**

38) **Bile** assists in emulsification of fats

39) **Crop** • A sac-like expanded section in esophagus of some animals

40) Amino group is **highly toxic** • It is removed by animals in **three** forms: • Ammonia • Urea, Uric Acid

41) Increase in carbon dioxide levels in blood also cause **pH to fall**

42) Loop of Henle has central importance in producing **concentrated urine**

43) Duodenum • First, shorter section **(25 cm)**

44) Av node is also called as__**atrioventricular node**

45) **Effect of Cholecystokinin**__Also stimulates release of bile from gallbladder

46) **Loop of Henle** has central importance in producing concentrated urine, OR

47) High Urea concentration done by **Loop of Henle**

48) Reabsorption at PCT • Cells of proximal convoluted tubule reabsorb **60–70 %** volume of ultra-filtrate.

49) Atrial contraction provides only **30% volume** of total mammalian ventricular output

50) The **K+** selective channels remain open in the resting membrane. **OR** (In resting membrane potential which ions move?)

51) In Kidney waste material are collected in **Collecting Duct, Ascending limb of loop of Henle and Distal tubule.**

52) Vena cava open into **the right atrium** of heart

53) **Somatostatin** inhibits gastric and pancreatic secretions

54) A villus has a covering **of epithelial cells.**

55) The Cells of Villi is made of **Epithelium**

56) How many types of Regulators___**3**.(**Types** • Osmoregulators • Oxyregulators • Thermoregulators)

57) Sac-like Gastrovascular Cavity is the Characteristic feature of **Cnidarians**

58) Bile salts are essential for the__**digestion of fats**

59) Proenzyme is also known as__**Zymogen**

60) Which disease is not Animal Models for Human Diseases?___**Alzhemier Disease**

61) **Pacemaker** • Any cardiac cells that can generate an impulse and can maintain heart rate.

62) Pacemaker's electrical activity spreads over the entire heart via **gap junctions.**

63) Most proteolytic enzymes are synthesized, stored, and released in an inactive molecular form: proenzyme or **zymogen**

64) The cells of adrenal medulla are modified **postganglionic sympathetic** neurons

65) RAAS is present in a specialized tissue: juxtaglomerular apparatus (JGA), located near the **afferent** arteriole

66) Atrial contraction provides only **30%** volume of total mammalian ventricular output

67) Diving animals are subjected to periods of **hypoxia and anoxia** during submergence

68) **Tropic hormones** act on other endocrine tissues and regulate their secretory activity.

69) Adenohypophysis secretes **four** tropic hormones:

70) **Neurofibrils** • Delicate threads running from cell body into the axon and dendrites.

Short & Long Questions…

1) Types of Feedback mechanism and explain negative feedback

Two types of regulatory feedback systems are present:

1. Negative feedback systems
2. Positive feedback systems

1. Negative feedback systems:

In life, the most common form of regulation encountered is the negative feedback, in which accumulation of an end product works to stop or slow down that process.

Example-1

The breakdown of sugar in the cells generates chemical energy in the form of ATP. When a cell makes more ATP than it can use, the excess ATP "feeds back" and inhibits an enzyme near the beginning of the pathway. This results in temporary stoppage of ATP production.

2) The factors of ultra-filtration?

Filtration takes place entirely passively across three layered wall of the glomerulus into the capsular space.

Filtration depends on three factors:

1. Net hydrostatic pressure
2. Colloidal osmotic pressure
3. Permeability of filtration membrane

3) Crypts of Lieberkühn

• Crypts of Lieberkuhn (mucosal glands) are dispersed throughout intestinal folds

• Produce a thinner, enzyme-rich alkaline fluid

• This fluid mixes with duodenal secretions

4) Importance pH regulation

PH regulation is the processes operating in living organisms to regulate the concentration of hydrogen ions and preserve a variable acid-base state.

Regulation of body fluid pH is one of the most important physiological functions at homeostasis, because activity of most chemical reactions via enzyme proteomes is dependent on fluid PH

PH of blood is maintained within a narrow alkaline range 7.35 to 7.45.

Outside this range pH becomes incompatible with life. Enzymes and proteins are denatured. Kidneys maintain PH by two ways:

- Reabsorption of bicarbonate ions
- Removal of fixed acids and H+.

5) Explain Blood vessels

Three types of blood vessels carry blood during circulation:

I. **Arteries** which carry blood away from the heart to organs and tissues.
II. **Veins** which return blood to the heart.
III. **Capillaries** which have thin walls formed of only a single layer of endothelial cells

and permit exchange of materials between blood and tissues.

6) Explain peristalsis and reverse peristalsis

Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract. It starts in the esophagus where strong wave-like motions of the smooth muscle move balls of swallowed food to the stomach.

Retro peristalsis is the reverse of the involuntary smooth muscle contractions of peristalsis. It usually occurs as a precursor to vomiting.

Local irritation of the stomach, such as bacteria or food poisoning, activates the emetic center of the brain which in turn signals an imminent vomiting reflex.

7) Explain Tubular reabsorption

Approximately 99% of the water and most of the useful substances are selectively reabsorbed from the ultra-filtrate in the tubular part of the nephron into the blood in surrounding peritubular capillaries.

The major reabsorbed molecules include glucose, amino acids, vitamins and inorganic salts. Due to reabsorption, the filtrate is converted into urine that has only left over nitrogenous and other waste substances when it leaves the tubule.

8) Tubular reabsorption at DCT

• Only 15–20 % of initial filtrate volume reaches DCT

• Tubular cells of DCT, actively reabsorb Na+ and Cl-

• Distal portions of DCT also contain ion pumps that reabsorb tubular Na+ in exchange for K+

• DCT also reabsorbs HCO3 - ions to regulate urine pH

9) Define Donnan Equilibrium

In 1911 Frederick Donnan discovered the phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane. This phenomenon is known as Donnan equilibrium and can be defined as:

“If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become

unequally distributed between the two compartments".

10) Define Donnan Effect

Donnan equilibrium implies that: if there is a no diffusible solute in one of the sol. separated by semi-permeable membrane, the conc. of sol. on both sides do not equalize. Conc. of sol. with non-diffusible solute remains high even at equilibrium. This is Donnan effect

11) Name the Constitute of lungs or Respiratory parts of lungs

Respiratory portion of the lung constitutes

- Terminal bronchioles
- Respiratory bronchioles
- Alveolar ducts
- Alveolar sacs

12) Uses of kymograph

- Study of skeletal muscle contractions (twitch and tetanus)
- Cardiac muscle activities (cardiac cycle)
- Measurement of blood pressure
- Rate of respiration

13) Define motility and its significant

The ability of the alimentary tract to contract and propel ingested material along its length is called motility.

The motility depends on the activity of smooth muscle walls and/or ciliary lining of the digestive tract. Motility of alimentary canal is important for many digestive functions to happen.

It is significant for:

- Translocation and conduction of food
- Mechanical treatment of food
- Prevent scumming.

14) Describe the absorptive Surface in small intestine

Epithelial lining of intestine is specialized for absorptive functions. It has:

- Increased surface area for absorption
- Rich supply of transport system

The adaptations to increase Surface area art enhanced by anatomical features:

Intestinal folds, Villi, Epithelial Cells of Villi, Microvilli Intestinal folds, Villi and microvilli collectively increase the surface area more than 500 times.

The Total area for absorption in small intestine of human is 200-300 m2

Comparison: gross cylindrical area of small intestine is only 0.4 m2.

15) Ornithine Urea cycle?

Ornithine-Urea Cycle is used by all ‘'vertebrates except teleost fishes. it Synthesize urea primarily in the liver. Two ammonia and one CO2 molecule are added to ornithine to form Arginine.

Enzyme arginase cleaves arginine to remove a molecule of urea and regenerate ornithine.

16) Note on lipases

Lipases are Lipid and fat digesting enzymes. Lipases Secreted in pancreatic juice and by

intestinal epithelium and Act on emulsified fat droplets and hydrolyze them into:

- Monoglycerides
- Diglycerides
- Free fatty acids and glycerol.

Emulsification by bile salts is necessary for lipases to hydrolyze lipids

- If bile salts are insufficient, fats are not emulsified, Fat digestion by lipases remains incomplete.

These are enzymes that the body uses to break down fats in food so they can be absorbed in the intestines. Lipase is produced in the pancreas, mouth, and stomach. For example Candida rugosa.

17) Myogenic pacemaker

• Consists of specialized self-excitatory muscle cells.

• Found in some invertebrate & all vertebrate hearts.

• Such hearts are known as myogenic hearts.

• In fishes and amphibians, situated in sinus venosus (a chamber of heart).

• In amniotes, located in superior posterolateral wall of the right atrium (a vestigial remnant of sinus venosus).

• Named as sinus node or sinoatrial node (SA node).

18) Ciliary motility

• Mechanism used by alimentary canals of annelids, many mollusks, tunicates, and cephalo-chordates. Involves beating of cilia lining the digestive tract

• Cilia generate fluid currents

• Fluid current propels food in digestive tube

19) What is myofibrils?

Within each muscle fiber, there are many myofibrils running in parallel fashion. Myofibrils are 1-2 µm in diameter and extend the entire length of the cell.

20) Appendix

The appendix is a narrow, finger-shaped pouch that projects out from the colon. Appendicitis occurs when the appendix becomes inflamed and filled with pus.

Appendicitis is an inflammation of the appendix, a finger-shaped pouch that projects from your colon on the lower right side of your abdomen.

21) Action potential

Definition

• Action potentials are the type of signals that neurons use to send information over long distances.

• Actions potentials are large but brief changes in membrane potential that are propagated along axons without decrement.

Importance of Action Potentials

• APs are responsible for every sensation, every memory, every thought — indeed every impulse.

They control:

• Effector responses

• Activation of voltage-gated ion channels

• Muscle contraction

• Exocytosis

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22) How Acetylcholine acts as slow neurotransmitters.

The most familiar fast acting neurotransmitter substance is the Acetylcholine. In most instances, acetylcholine has an excitatory effect.

It also acts as inhibitory transmitter in some instances e.g. at the peripheral parasympathetic nerve endings where it is involved in the inhibition of the heart by the vagus nerves.

Mode of Action of Acetylcholine

- When acetylcholine (ACh) is released into the synaptic cleft, it binds to the ligand-gated ACh specific receptors in the postsynaptic membrane.
- The binding causes Na+ and K+ ion channels to open briefly that produce an EPSP.

23) Transverse tubule

T Tubules

T tubules are thin internal extensions of the cell membrane. They are less than 0.1 µm in diameter. They innervate the cell at the level of Z disk. Make branching networks around the perimeter of each myofibril.

Role of T Tubules

• T tubule system provides the anatomic link between the surface membrane and the myofibrils deep inside the muscle fiber. When an action potential spreads over a muscle fiber membrane, a potential change also spreads along the T tubules to the deep interior of the muscle fiber.

• In the cell interior, T tubes are linked to SR.

• SR releases Ca2+ that permit myosin cross-bridges to attach and generate force for contraction.

24) Discuss the function of triad (3)

The triad is a skeletal muscle substructure responsible for the regulation of excitation-contraction coupling. It is formed by the close apposition of the T-tubule and the terminal sarcoplasmic reticulum.

25) Acclimatization or acclimation effect and examples (5)

Acclimatization

• An adaptive change within an individual animal. • Happens due to chronic exposure to new, naturally occurring environmental conditions.

Acclimatization Example

- An animal migrates to a high altitude area
- Faces low oxygen partial pressure

Effects

- Lung ventilation rate increases initially
- Ventilation drops to normal in few days
- Animal has acclimatized to the high-altitude

Acclimation

- Changes are induced experimentally in the laboratory

Acclimation Example

- Simulating high-altitude conditions in lab.
- An animal physiologist places the animal in a hypobaric chamber.

Effects

• Breathing rate reacts in the same way.

• Animal becomes acclimated to the experimental conditions.

26) Characteristics Of hormones

• Hormones coordinate long-term functions of animal tissues and organs. e.g., growth and maintenance, sexual activity, reproductive cycles, modification of behavior.

• They perform many regulatory functions. e.g., osmoregulation, Regulation of blood sugar levels, control of metabolic rate

• Many rhythmic activities of animals are due to hormones. e.g. Molting, Sleep-wake up cycles, Hunger, Seasonal Reproductive activations, Migration in birds

- Maintain homeostasis
- Coordinate body's responses to stress
- Mediate responses to many environmental stimuli.

27) Give Detail about Homeostasis With Other Examples

Defination:

- Walter Cannon (1929)
- The tendency of an organism to maintain relative internal stability
- To keep internal fluctuations in a narrow range.
- “The ability to protect internal environment from the harms of fluctuations in external environment. Homeostasis does not mean to keep a fixed internal environment.”

Other Examples of Homeostasis

- Homeostatic regulation of water
- Homeostatic regulation of temperature
- Regulation of pH
- Regulation of glucose concentration
- Regulation of osmotic pressure
- Regulation of oxygen level
- Regulation of ion conc.

28) Write note on Urea and Uric acid?

Urea

Terrestrial animals convert ammonia into urea or uric acid which can be excreted with less water loss. Urea is less toxic than ammonia. It requires only 0.05 L of water to excrete 1 g of urea (10% as compared to ammonia)

• Urea synthesis is energetically costly as it consumes ATP

Uric Acid

Excretion of uric acid requires least water i.e. only 0.001 L for 1 g uric acid (1% as compared to ammonia). It is only slightly soluble in water. It is excreted as a white pasty precipitate

29) Neurofibrils:

Delicate threads running from cell body into the axon and dendrites.

Neurofibrils— Functions

• Give support and shape to the neuron.

• Transport substances from cell body to the distal cell processes.

30).End_Diastolic Volume Four Steps:

- Venous filling pressure
- Pressures generated during atrial contraction
- Distensibility of the ventricular wall
- Time available for filling the ventricle

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Remember me in you Precious Prayer

Jazak Allah